



GERALD R. FORD SCHOOL
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Public Policy and Technology Governance

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Technology is not neutral.

Other lesser known facts about technology

- Technology is not separate from society or politics
- Innovation is not always a public good
- We can anticipate the implications of technologies, and can shape its development through regulation

Technology, meet society

- Technology and technology policy both reflect and reinforce social, political, and moral orderings
 - What we choose to invest in
 - How and how much we choose to invest
 - How we build, distribute, and implement
 - Who designs it, who it is designed for

Case example: Facial recognition

Eight Months Pregnant and Arrested After False Facial Recognition Match

Porcha Woodruff thought the police who showed up at her door to arrest her for carjacking were joking. She is the first woman known to be wrongfully accused as a result of facial recognition technology.

February, 2023



For all mankind?

- Innovation is not always a public good
- If a new technology is supposed to be for the public good, ask yourself, “Which public?”
 - Who, specifically, benefits from this tech, and who might not?
 - Who is responsible?

Stakeholders

- Organized
 - Companies
 - Advocacy groups
 - Government agencies
- Unorganized (Who *might* have an interest in a particular technology?)
 - Neighbors
 - Potential users
 - Patients
 - Students

Case example: Traffic light optimization

Even if they're perfectly “accurate,” these systems still have social consequences.



So predictable

- All new technologies, no matter how cutting edge, have things in common with past technologies
- We can look to past technologies to make predictions about the likely implications of new technology

Case example: Generative AI

IMPLICATIONS OF LLM DEVELOPMENT

1

LLMs WILL EXACERBATE **ENVIRONMENTAL INJUSTICE**

LLM processing occurs in physical data centers, which require massive amounts of natural resources. Data center construction is already disproportionately harming marginalized populations.

2

LLMs WILL ACCELERATE **THE THIRST FOR DATA**

To operate properly, LLMs consume enormous amounts of data. This will challenge traditional ideas of individual informed consent and create incentives for unethical data extraction and use.

3

LLMs WILL BE DEPLOYED INITIALLY AS **ASSISTIVE**

To gain social acceptance, developers will initially introduce LLMs into large, authoritative systems. They will also emphasize the technology's modularity and capacity to empower users.

IMPLICATIONS OF LLM ADOPTION

4

LLMs WILL REINFORCE **INEQUALITY**

Neither the development landscape nor LLM datasets represent marginalized communities adequately. As a result, we expect LLMs to systematically minimize and misrepresent these voices while amplifying the perspectives of the already powerful.

5

LLMs WILL REMAKE **LABOR AND EXPERTISE**

LLMs will transform, rather than replace, many occupations. Humans will shift to more complex and risky tasks, and many workers and consumers will resist these technologies.

6

LLMs WILL REINFORCE **SOCIAL FRAGMENTATION**

LLMs will help communities gather information that aligns with their interests and values. This may democratize knowledge, but it will also challenge expert authority and erode shared realities.

Ask the experts

Many different types of expertise are relevant in the development and governance of technology, technical expertise is just one of them.

- History
- Policy
- Sociology
- Linguistics
- Lived experience
- Relationships
- Design
- Community engagement
- Psychology
- Economics



Join by Web PollEv.com/davidwentzloff730



Think about some technologies or apps you enjoy using, like Tik Tok or ChatGPT. What are some reasons you would stop using them?

Nobody has responded yet.

Hang tight! Responses are coming in.

Technology and policy

Practicalities of tech policy

Levels of policy

- Federal and state constitutions
- Federal and state statutes
- Case law
- Contracts
- International treaties

Players

- Industry lobbyists
- Government relations
- Lawmakers and legislative staffers
- Advocacy groups
- Academia

Many kinds of policy touch tech

- Environmental regulation
- Labor law
- Tax law
- Constitutional rights
- Intellectual property
- Disability rights
- Immigration policy
- Innovation policy
- Economic policy
- Housing policy
- Transportation policy
- International diplomacy
- Privacy policy
- Education policy

Tik Tok vs. Privacy

- The U.S. has no federal privacy law.
- Most social media companies collect user data, and so do credit card companies, and advertisers, and retailers, and...



Think/pair/share

People often trade their privacy for convenience, access, or discounts.

What are some risks of giving corporations access to personal information about yourself?

Are there situations where you would not be willing to share information with a company? What are they?

Thirst for data vs. Environment

Data centers fuel AI and crypto but could threaten climate, experts say

Data centers' energy usage is expected to soar in the coming years, experts say.

By [Max Zahn](#)

April 20, 2024, 6:08 AM



Automation vs. equity



Chicago's "Race-Neutral" Traffic Cameras Ticket Black and Latino Drivers the Most

January, 2022

Talent pipeline vs. immigration policy



- Changes to visa policies can disrupt hiring and create labor shortages in specialized tech sectors
- On a macro scale, can affect global competitiveness.

What is the Science, Technology, & Public Policy Program?

- Research center based in the Ford School of Public Policy
- Affiliated UM faculty from sciences, engineering, and professional schools.
- Educational, research, and community & policy engagement elements
- Science & technology broadly defined, everything from AI to biotech to environment

What is the Science, Technology, & Public Policy Program? continued

- Education

- Graduate Certificate Program
- Massive Open Online Course
- Expanding undergraduate & postdoctoral opportunities

- Research

- Technology Assessment Project
- Student Research Corps

- Policy Engagement

- Community Partnerships Initiative



Questions?

Attendance Answer

Mo Bamba

Thanks for a great semester!